Chapter-5

Bernoulli Distribution

Bernoulli distribution is a discrete probability distribution for a Bernouli trial- a random experiment that has only two outcomes

success - 1

failure - 0

the probability density function for this distribution is

$$f(x) = \begin{cases} p & \text{if } x=1 \\ q & \text{if } x=0 \end{cases}$$

which also can be written as

$$f(x) = p^x q^{1-x} \quad x=0,1$$

Mean:

$$E(x) = \sum_{x=0}^{1} x f(x)$$

$$= \sum_{x=0}^{1} x p^x q^{1-x}$$

$$= 0. + 1. p^1 q^{1-1} = p$$

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$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \sum_{x=0}^{1} x^2 p^x q^{1-x}$$

$$= 0 + 1\, p\, q^0$$

$$= p$$

$$V(x) = p - p^2$$

$$= p(1-p)$$

$$= pq$$

$$\underbrace{SSS\text{ ---} S}_{x}\ \underbrace{FFFF\text{ --} F}_{n-x}$$

$$P(S) \rightarrow p$$
$$P(F) \rightarrow q$$

$$P[SSS\text{ ---} SFFFF\text{ ---} F]$$

$$= [ppp\text{ ---} pqqqq\text{ ---} q]$$

$$= p^x q^{n-x}$$

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Binomial Distribution

The random variable x that denotes the number of success in n Bernoulli trials has a binomial distribution

$$f(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0 \dots n$$

$$f(x; n; p) = 0 \quad \text{otherwise}$$

$f(x)$ ~~=~~ f where n is positive integer
p and $0 \leq p \leq 1$

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Show the sum of $\sum f(x; n, p) = 1$ where $x \sim (n, p)$

$$f(x) = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x}$$

$$= (p+q)^n$$

$$= 1^n = 1$$

Mean, variance,

$$E(x) = \sum_{x=0}^{n} x \, f(x)$$

$$= \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

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$$= \sum_{x=0}^{n} x \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} \frac{n(n-1)!}{(x-1)!(n-1-(x-1))!} p^{x-1} q^{n-x}$$

$$= \sum_{x=1}^{n} p \cdot \frac{n(n-1)!}{(x-1)!(n-1-(x-1))!} p^{x-1} q^{n-x}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{n-x}$$

letting $y = x-1$ $\quad x=1 \Rightarrow x-1=0$ $\quad x \to n,\ y \to n-1$

$$E(X) = np \sum_{y=0}^{n-1} \frac{(n-1)!}{y!(n-1-y)!} p^y q^{n-1-y}$$

$$= np\,(p+q)^{n-1}$$

$$= np$$

* Show the mean in binomial distribution is greater than its varciance.

$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = E(x(x-1)+x)$$

$$= E[x(x-1)] + E[x]$$

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$$E[x(x-1)] = \sum_{x=0}^{n} x(x-1) \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} q^{n-x}$$

$$= n(n-1)p^2 \sum_{Y=0}^{n-2} \frac{(n-2)!}{Y!(n-2-Y)!} p^Y q^{n-2-Y}$$

let $Y = x-2$
$x \to n$
$Y \to n-2$

$$= n(n-1)p^2 (p+q)^{n-2}$$

$$= n(n-1)p^2 (1)^{n-2}$$

$$= n(n-1)p^2$$

$$\therefore E[x^2] = n(n-1)p^2 + np$$

$$V(x) = n(n-1)p^2 + np - n^2p^2$$

$$= n^2p^2 - np^2 + np - n^2p^2$$

$$= np - np^2$$

$$= np(1-p)$$

$$= npq$$

[note: $q = 1-p$ so $npq < np$]

∴ Mean is greater than variance.

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Moment generating function.

MGF, $M_x(t) = E(e^{tx})$

$= (pe^t + q)^n$

Poisson Distribution

$\sum m(x) = 1$

$$f(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!} \qquad x = 0, 1, 2 \cdots$$

$= 0$ otherwise

* the parameter λ is called the mean arrival rate or mean occurrence rate.

$$\sum_{x=0}^{\infty} f(x;\lambda) = \sum_{x=0}^{\infty} \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$= e^{-\lambda} e^{\lambda}$

$= 1$

$\left[e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots\right.$

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Mean -

$$E(x) = \sum_{x=0}^{\infty} x f(x)$$

$$= \sum_{0}^{n} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{n} \frac{x \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=1}^{n} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= e^{-\lambda} \lambda \left[\frac{1}{0!} + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots \right]$$

$$= e^{-\lambda} \lambda e^{\lambda}$$

$$= \lambda$$

$$E(x^2) = \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x^2 \lambda^x}{x!}$$

$$= \lambda^2 + \lambda$$

$$V(x) = \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$Mx(t) = e^{\lambda(e^t - 1)}$$

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*** Find MGF, Mean, median of binomial distribution and show ~~ne~~ mean is greater than variance.

the random variable x that denotes the number of success in a Bernouli trials has a binomial distribution given by

$$f(x;n,p) = \binom{n}{x} p^x q^{n-x} \quad x = 0, \sim\sim 1$$

where n is positive integer and $0 \le p \le 1$

Moment generating function

$$M_x(t) = E(e^{tx})$$

$$= (pe^t + q)^n$$

Mean.

$$E(x) = \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=0}^{n} x \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

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$$= \sum_{x=1}^{n} \frac{n(n-1)!}{(x-1)!\,(n-1-(x-1))!}\, p\, p^{x-1}\, q^{n-x}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!\,(n-1-(x-1))!}\, p^{x-1}\, q^{n-x}$$

letting $y = x - 1$

$$E(x) = np \sum_{y}^{n-1} \frac{(n-1)!}{y!\,(n-1-y)!}\, p^{y}\, q^{n-1-y}$$

$$= np\,(p+q)^{n-1}$$

$$= np$$

There is no single formula to find median for a binomial distribution. However if the value for np is integer or whole number, the mean, median and mode for all equal np.

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variance

$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = E\left(x(x-1) + x\right)$$

$$E(x(x-1)) = \sum_{x=0}^{n} x(x-1) \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \frac{(n-2)!}{(x-2)!(n-2-(x-2)!} p^{x-2} q^{n-x}$$

letting $x-2 = y$

$$= n(n-1)p^2 \sum_{y=0}^{n-2} \frac{(n-2)^2}{y!(n-2-y)!} p^y p q^{n-2-y}$$

$$= np^2(n-1)\ (p+q)^{n-2}$$

$$= (n^2p^2 - np^2)(1)$$

$$E(x^2) = n^2p^2 - np^2 + np$$

$$V(x) = n^2p^2 - np^2 + np - n^2p^2$$

$$= np(1-p)$$

$$= npq$$

as $npq < np$ so mean is greater than variance.

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v^I

What do you mean by binomial variate? suppose x follows Binomial distribution with parameters n and p.

Show

i) $E\left(\frac{x}{n}-p\right)^2 = \frac{pq}{n}$

ii) $cov\left[\frac{x}{n} - \frac{n-x}{n}\right] = \frac{-pq}{n}$

i) $E\left(\frac{x}{n}-p\right)^2 = E\left(\frac{x^2}{n^2} - 2\frac{xp}{n} + p^2\right)$

$= \frac{1}{n^2} E(x^2) - \frac{2p}{n} E(x) + E(p^2)$

$= \frac{1}{n^2}\left(n(n-1)p^2 + np\right) - \frac{2p}{n} np + p^2$

Here $E(b) = b$, $E(x) = np$ $E(x^2) = n(n-1)p^2 + np$

$= \frac{1}{n^2}\left(n^2p^2 - np^2 + np\right) - 2p^2 + p^2$

$= p^2 - \frac{p^2}{n} + \frac{p}{n} - p^2$

$= \frac{p}{n}(1-p)$

$= \frac{pq}{n}$

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ii) $\text{cov}[X,Y] = E(X,Y) - E[X] \cdot E[Y]$

$\therefore \text{cov}\left[\frac{x}{n}, \frac{n-x}{n}\right] = E\left(\frac{x}{n} \cdot \frac{n-x}{n}\right) - E\left(\frac{x}{n}\right) E\left(\frac{n-x}{n}\right)$

$= \frac{1}{n^2} E\left(x(n-x)\right) - \frac{1}{n} E(x) \cdot \frac{1}{n} E(n-x)$

$= \frac{1}{n^2} E\left(xn - x^2\right) - \frac{np}{n} \cdot \frac{1}{n}\left[E(n) - E(x)\right]$

$= \frac{1}{n^2} E(xn) - \frac{1}{n^2} E(x^2) - \frac{p}{n}(np - np)$

$= \frac{n}{n^2}(np) - \frac{1}{n^2}\left(n(n-1)p^2 + np\right) - p + np^2$

$= p - \frac{1}{n^2}\left(n^2p^2 - np^2 + np\right) - p + np^2$

$= -p^2 + \frac{p^2}{n} - \frac{p}{n} + np^2$

$= \frac{-np^2 + p^2 - p + n^2p^2}{n}$

$= \frac{-p(1-p)}{n}$

$= \frac{-pq}{n}$

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A particulare telephone numbere is used to receive both calls and fax messages. suppose that 25% of the incoming calls involbe fax msg and consider a sample of 25 incoming calls.

What is the probability that

i) at most six of the calls involve a fax msg

2) exactly six of the calls involve a fax msg.

3) at least six " "

4) more than six " "

here $n = 25$, x be the calls involve a fax msg

$p = 25\% = 0.25$

$q = 1 - p = 0.75$

$$f(x; n, p) = \binom{n}{x} p^x q^{n-x} \qquad x = 0 \text{ - - - } n$$

i) $$P(x \le 6) = \sum_{x=0}^{6} f(x)$$

$$= \sum_{x=0}^{6} \binom{25}{x} p^x q^{25-x}$$

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$$= \binom{25}{0} p^0 q^{25} + \binom{25}{1} (p) q^{24} + \binom{25}{2} p^2 q^{23} +$$

$$\binom{25}{3} p^3 q^{22} + \binom{25}{4} p^4 q^{21} + \binom{25}{5} p^5 q^{20}$$

$$+ \binom{25}{6} p^6 q^{19}$$

$$= 0.561098$$

ii) $P(x=6) = \binom{25}{6} 0.25^6 q^{25-6}$

$$= 0.1828$$

iii) $P(x \geq 6)$

$= 1 - P(x \leq 6)$

$= 1 - \{ P(x \leq 6) - P(x=6) \}$

$= 1 - (0.561098 - 0.1828)$

$= 1 - 0.3783$

$= 0.62172$

iv) $P(x > 6)$

$= 1 - P(x \leq 6)$

$= 1 - 0.56109$

$= 0.48389$

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The geometric Distribution

X be the number of trials required to achieve the first success. The distribution of x is given by

$$P(x) = q^{x-1} p \quad , \quad x = 1, 2 \ldots$$

$$= 0 \qquad \text{otherwise}$$

It is easy to ~~ver~~ verify that this a ~~prob~~ Pdf since

$$\sum_{x=1}^{\infty} pq^{x-1} = p \sum_{x=1}^{\infty} q^{x-1}$$

let $x-1 = Y$

$$= p \sum_{Y=0}^{\infty} q^{Y}$$

$$= p \left[\frac{1}{1-q}\right]$$

$$= 1$$

$P(x) \geq 0$ for all x

$$\text{Mean} = \frac{1}{p} = \mu = E(x)$$

$$\sigma^2 = \frac{q}{p^2} = V(x)$$

$$M_x(t) = \frac{pe^t}{1 - qe^t}$$

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2) ... order but different ... whenever observations are not only ... ranked

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Lecture - 10

chapter - 05

1) $(a+b)^n = \binom{n}{0} a^{n-0} b^0 + \binom{n}{1} a^{n-1} b^1 + \cdots + \binom{n}{n} a^{n-n} b^n$

$$= \sum_{x=0}^{n} \binom{n}{x} a^{n-x} b^x$$

$$(a+b)^{n-1} = \sum_{x=0}^{n-1} \binom{n-1}{x} a^{n-1-x} b^x$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Random experiment

- success - 1
- failure - 0

$$f(x) = \begin{cases} p & \text{if } x = 1 \\ q & \text{if } x = 0 \end{cases}$$

$$f(x) = p^x q^{x-1}, \quad x = 0, 1$$

this is called Bernoulli Distribution.

$$E(x) = \sum_{x=0}^{1} x f(x)$$

$$= \sum_{x=0}^{1} x\, p^x q^{x-1}$$

$$= 1 \cdot p^1 q^{1-1} = p$$

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$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \sum_{x=0}^{1} x^2 \; p^x \; q^{1-x}$$

$$= 1^2 \; p \; q^0 = p$$

$$V(x) = p - p^2$$

$$= p(1-p)$$

$$= pq$$

S S S S - - - S F F F - - F

x $\quad$ $n-x$

success $\quad$ fail

$$P[SSSS \text{---} SFFF \text{---} F]$$

$$= pppp \ldots pqqq \ldots q$$

$$= p^x \; q^{n-x}$$

$$\left[f(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0 \text{---} n \right)$$

$$\downarrow$$

$$f(x : n, p)$$

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#

Show that sum of $\sum f(x:n,p) = 1$ where $x \sim$ Binomial (n,p)

binomial distribution- $f(x) = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x}$

$$= (p+q)^n = 1^n = 1$$

$$\Rightarrow E(x) = \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=0}^{n} x \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} \frac{n(n-1)}{(x-1)(n-1-(x-1))!} p^{x-1} q^{n-x}.$$

$$= np \sum_{x-1=0}^{n-1} \frac{(n-1)!}{(x-1)!((n-1)-(x-1))!} p^{x-1} q^{n-1}$$

Let $y = x-1$ $= np \sum_{y=0}^{n-1} \frac{(n-1)!}{y!(n-1-y)!} p^y q^{n-1-y}$

$$= np(q+q)^{n-1}$$

$$= np$$

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Show that the mean of binomial distribution is greater than its variance.

$\underline{V = npq}$

$E(x^2) = E(x(x-1) + x)$

$= E(x(x-1)) + E(x)$

$E(x(x-1)) = \sum_{x=0}^{n} x(x-1) \frac{n!}{x!(n-x)!} p^x q^{n-x}$

$= n(n-1)p^2 \sum_{x=2}^{n} \frac{(n-2)!}{x(x-1)(n-2)!(n-x)!} p^{x-2} q^{n-x}$

$= n(n-1)p^2 \sum_{y=0}^{n-2} \frac{(n-2)!}{y!(n-y-2)!} p^y q^{n-y-2}$

$= n(n-1)p^2$

$E(x^2) = n(n-1)p^2 + np$

$V(x) = n^2p^2 - np^2 + np - n^2p^2$

$= np(1-p)$

$= npq$

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binomial distribution -

$$P(x) = \binom{n}{x} p^x q^{n-x} \qquad x = 0, 1, 2 \cdots n$$

$$= 0$$

$P(x$ success in n^{th} trials)

the

Moment generating function.

$$MGF = E(e^{tx})$$

$$= (pe^t + q)^n$$

Poison's Distribution -

sum of $m(x) = 1$

$$f(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!} \qquad x = 0 \cdots \infty$$

$$\sum_{x=0}^{\infty} f(x;\lambda) = \sum_{x=0}^{\infty} \frac{e^{-\lambda}\lambda x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} e^{\lambda}$$

$$= 1$$

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Mean. $$E(X) = \sum_0^n x f(x)$$

$$= \sum_0^n x \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_0^n x \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_1^n \frac{\lambda^{x-1}}{(x-1)!}$$

$$= e^{-\lambda} e^{\lambda} \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right]$$

$$= e^{-\lambda} \lambda e^{\lambda}$$

$$= \lambda$$

$$E(X^2) = \sum_0^{\infty} x^2 \cdot \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= \lambda^2 + \lambda$$

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Lecture - 10

Q What do you mean by Binomial variate? Suppose X follows Binomial distribution with parameter n and p show that

(i) $E\left[\frac{x}{n} - p\right]^2 = \frac{pq}{n}$

(ii) $cov\left[\frac{x}{n}, \left(\frac{n-x}{n}\right)\right] = -\frac{pq}{n}$

(i) $E\left[\frac{x}{n} - p\right]^2 = E\left[\frac{x^2}{n^2} - 2\frac{x}{n}p + p^2\right]$

$= \frac{1}{n^2}E[X^2] - 2\frac{p}{n}E[X] + E[p^2]$

$= \frac{1}{n^2}E[X(X-1) + X] - 2\frac{p}{n}np + p^2$

$= \frac{1}{n^2}\left[E[X(X-1)] + E[X]\right] - 2p^2 + p^2$

$= \frac{1}{n^2}\left[n(n-1)p^2 + np\right] - p^2$

$= \frac{(n^2-n)p^2}{n^2} + \frac{p}{n} - p^2$

$= \frac{n^2p^2 - p^2n + np - n^2p^2}{n^2}$

$= \frac{np - p^2n}{n^2}$

$= \frac{p(1-p)}{n}$

$= \frac{pq}{n}$

Ans.

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(2)

ii) $$\text{Cov}[X,Y] = E[X.Y] - E[X] \cdot E[Y]$$

$$\text{Cov}\left[\frac{x}{n}, \frac{n-x}{n}\right] = E\left[\frac{x}{n} \cdot \frac{n-x}{n}\right] - E\left[\frac{x}{n}\right] \cdot E\left[\frac{n-x}{n}\right]$$

$$= \frac{1}{n^2} E[x(n-x)] - \frac{1}{n} E[x] \cdot \frac{1}{n} E(n-x)$$

$$= \frac{1}{n^2} E(nx - x^2) - \frac{1}{n^2} E(x)\, E(n-x)$$

$$= \frac{1}{n^2}\left[nE(x) - E(x^2)\right] - \frac{1}{n^2} np\left[E(n) - E(x)\right]$$

$$= \frac{1}{n^2}\left[n \cdot np - n(n-1)p^2 - np\right] - \frac{1}{n^2} np(n-np)$$

$$= \frac{n^2p}{n^2} - \frac{n^2p}{n^2} + \frac{np^2}{n^2} - \frac{np}{n^2} - \frac{n^2p}{n^2} - \frac{n^2p^2}{n^2}$$

$$= \frac{p^2}{n} - \frac{p}{n} - p - p^2$$

$$= \frac{p(p-1)}{n} - p(p-p^2)$$

$$= \frac{-p(1-p)}{n} - pq$$

$$= \frac{-pq}{n} + pq$$

$$= \frac{1}{n^2}\left(n^2p - n^2p^2 + np^2 - np\right) - p(1-p)$$

$$= p - p^2 + \frac{p^2}{n} - \frac{p}{n} - pq$$

$$= -\frac{p(1-p)}{n} + p(1-p) - pq$$

$$= \frac{-pq}{n} + pq - pq = \frac{-pq}{n}$$

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Q. A particulart telephone numbere is used to receive both voice calls and fax messages. Suppose that 25% of the incoming calls involve fax messages and considere a sample of 25 incoming calls.

What is the probability that -

i) at most six of the calls involve a fax messages.

ii) Exactly six of the calls involve a fax messages.

iii) at least six of the calls involve a fax messages.

iv) more that six of the calls involve a fax messages.

$n = 25$

$p = 25\% = 0.25$

$f(x; n, p) = \binom{n}{x} p^x q^{n-x}$ $\quad x = 0 \cdots n$

let x be the calls involves fax messages.

5

i) ~~Let x~~

i) $P(x \leq 6) = \sum_{x=0}^{6} f(x)$

$$= \sum_{x=0}^{6} \binom{25}{x} p^{x} q^{25-x}$$

$$= \sum_{x=0}^{6} \binom{25}{x} (.25)^{x} (0.75)^{25-x}$$

$$= \binom{25}{0} (0.25)^{0} (0.75)^{25} + \binom{25}{1} (0.25)^{1} (0.75)^{25-1}$$

$$+ \binom{25}{2} (0.25)^{2} (0.75)^{25-2} + \binom{25}{2} (0.25)^{3} (0.75)$$

ii) $P(x=6) = \binom{n}{6} \cdot p^6 q^{n-6}$

$= \binom{25}{6} (0.25)^6 q^{25-6}$

$=$

iii) $P[x \geq 6] = 1 - P(x \leq 5)$

iv) $P[x > 6] = 1 - P[x \leq 6]$

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Q1. what is PDF of Normal distribution and standard normal distribution? Show that for ~~a~~ normal distribution mean, median mode are ~~are~~ equal?

Q2. The article of "Monte Carlo Simulation - Tool for better understanding of ~~LRFD~~ LRFD" (J. Structural Engr. 1993: 1586 -1599) suggest that ~~yeild~~ ~~yeild~~ yield strength (ksi) for A36 steel grade is normally distributed with $\mu = 43$ and $\sigma = 4.50$.

what is the probability that yield strength is i) at most 40

ii) Greater than 60?

iii) Between 35 to 55.

2) ~~$P(X \leq 40) = P\left(\frac{X-\mu}{\Delta} < \frac{40-\mu}{\Delta}\right)$~~

$X \sim N(\mu, \Delta^2)$

$\mu = 43$

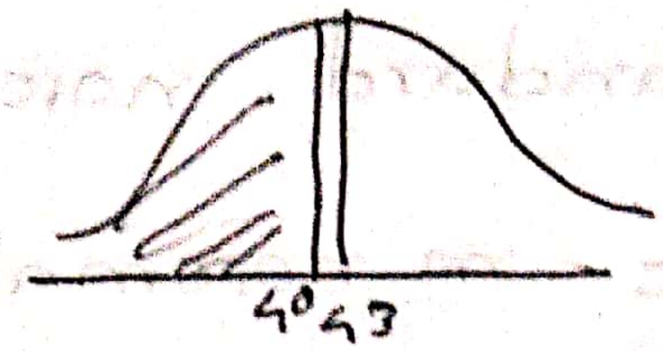
$x = 2.5$

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∞

i) $P(X \leq 40)$

$= P\left(\frac{X-\mu}{\Delta} \leq \frac{40-\mu}{\Delta}\right)$



$= P(Z \leq -0.66)$

$= \phi(-0.66)$

$= 0.25143$

ii) $P(X > 60) = 1 - P(X < 60)$

$= P\left(\frac{X-\mu}{\Delta} < \frac{60-43}{4.5}\right)$

$= P(Z < 3.77)$

$= \phi(3.78)$

$= 0.99992 \quad = 1 - 0.99992 = 0.00008$

iii)

$P[a < X < b]$

$= F(b) - F(a)$

$P(35 < X < 55)$

$= P\left(\frac{35-\mu}{\Delta} < \frac{X-\mu}{\Delta} < \frac{55-\mu}{\Delta}\right)$

$= P(-1.78 < Z < 2.67)$

$= \phi(2.67) - \phi(-1.78)$

$= 0.99621 - 0.03754$

$= 0.95867.$

The geometric Distribution (5.4)

$$P(x) = q^{x-1} p, \qquad x = 1, 2 \cdots$$

$$= 0 \qquad \text{otherwise.}$$

$$\text{Mean} = \frac{1}{p} = \mu = E[x]$$

$$\sigma^2 = \frac{q}{p^2} = V(x)$$

$$\text{MGF}, \quad M_x(t) = \frac{pe^t}{1 - qe^t}$$